

Dubai Real Estate Market Analysis

Power BI Portfolio Case Study

Executive Summary

Built an end-to-end Power BI solution to analyze Dubai real-estate pricing, rental performance, off-plan premiums, investment yield, market activity, and location/property characteristics. The project combines five source datasets into a dimensional semantic model and a five-page analytical report designed for fast business interpretation.

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SOURCE DATASETS	FACT TABLES	DIMENSIONS	RELATIONSHIPS	REPORT PAGES

Business Questions

- Is Dubai property pricing becoming more expensive over time?
- Are off-plan properties priced at a premium to secondary-market properties?
- Which communities are most expensive and which show stronger rental-yield potential?
- How do property type, bedroom count, property size, and accessibility relate to secondary-market property prices?

Tools & Approach

Power BI Desktop • Power Query • DAX • Dimensional modeling • Data profiling • Time intelligence • Interactive visualization

Market Definitions

Off Plan: Property marketed or sold before construction is complete and before final handover, typically by a developer. In this project it supports project, pricing, launch/handover, payment-plan, and price-per-sq-ft premium analysis.

Secondary Market: Property resold by an existing owner/investor rather than as the developer's initial sale. In this project Secondary Sales is the established resale-market pricing benchmark and the main source for Location & Property Analysis.

1. Data Preparation & Modeling

Source: Kaggle. The data contains realistically calibrated, modelled listing-level records with real geographic and market anchors. It is suitable for analytical demonstration but should not be interpreted as official Dubai Land Department transaction data.

Source Tables

Table	Role / Grain	Rows
Off Plan	Listing-level fact	12,000
Rentals	Listing-level fact	25,000
Secondary Sales	Listing-level fact	50,000
Area Prices Monthly	Community × month fact	6,384
Metro Stations	Reference source for station dimension	55

Key Preparation Decisions

- Profiled full datasets for Off Plan, Rentals, and Secondary Sales; retained important data-quality findings and standardized naming/data types.
- Standardized currency and price/rent fields, property labels, date fields, metro-distance naming, and location-related columns.
- Preserved nulls where analytically valid rather than forcing replacements; for example, Secondary Sales floor fields are ~36% empty.
- Retained physically required source columns where query dependencies existed, but hid redundant/technical fields from report view to keep the semantic layer clean.
- Final QA corrected Dim Community coverage by sourcing Community and Zone from Area Prices Monthly, expanding the dimension from 43 to all 84 communities and eliminating unmatched blank Community categories.

Semantic Model

A star-schema-oriented model was created with five conformed dimensions: Dim Date, Dim Community, Dim Project, Dim Metro Station, and Dim Bedrooms. The earlier Dim Location design was removed because it duplicated filtering responsibilities and complicated the shared model.

Dimension	Connected Facts
Dim Date	Off Plan, Rentals, Secondary Sales, Area Prices Monthly
Dim Community	Off Plan, Rentals, Secondary Sales, Area Prices Monthly
Dim Project	Off Plan
Dim Metro Station	Off Plan, Rentals, Secondary Sales
Dim Bedrooms	Rentals, Secondary Sales

All 14 relationships are active, many-to-one, and single-direction. No fact-to-fact relationships are used. The two added relationships connect Dim Bedrooms[Bedrooms] (1) to Rentals[Bedrooms] (*) and Secondary Sales[Bedrooms] (*), enabling one shared Bedroom Label slicer across both fact tables.

2. Analytical Logic

The measure layer focuses on reusable business metrics rather than visual-specific calculations. Temporary/debug measures were removed during final cleanup.

Off Plan Price per Sq Ft Premium %: Compares average off-plan PPSF with secondary PPSF in the current filter context.

Estimated Gross Rental Yield %: Average annual rent ÷ average secondary price in the current filter context. It is a market-level screening metric, not a matched-property return, because Rentals and Secondary Sales do not share a common property/unit ID.

Rental Listings YoY %: Uses same-period prior-year logic so partial 2026 is compared fairly with the equivalent 2025 period.

Average Annual Rent YoY %: Measures annual-rent change versus the previous-year date context.

Listing Share & Activity Measures: Track Off Plan, Secondary, and Rental market activity across time and segments. Rental Listing Records counts rental-table rows under the active filters and does not represent unique physical properties.

Average / Median Price & Rent: Provides both central-tendency views to reduce distortion from luxury-property outliers.

3. Report Structure

Page	Focus	Examples of Analysis
1. Market Overview	Market direction	Price/rent trends, financing rates, annual listing activity
2. Off Plan vs Secondary	Pricing comparison	PPSF premium, listing share, community/property-type comparison
3. Rental Market	Rental performance	Rent levels, annual rent PPSF, furnishing status, community comparison, YoY movement
4. Investment & Yield	Investment screening	Gross yield, price vs rent positioning, community opportunities
5. Location & Property Analysis	Secondary-market property & accessibility	Secondary price, metro travel time, bedrooms, size, decomposition tree

Design Standards

Continuous date axes were used for monthly trends to preserve chronology. Currency uses USD with consistent decimal formatting; rental PPSF is annual USD per sq ft; rate fields stored as percentage points remain decimal values; and count measures are whole numbers. Fact-table copies of dimension attributes were hidden from report view to guide users toward conformed dimensions. Location & Property Analysis uses Secondary Sales for its analytical values; measure Home-table placement is organizational only and does not determine DAX source logic.

4. Key Findings

\$1.70M AVG OFF PLAN PRICE	\$1.01M AVG SECONDARY PRICE	\$83.69K AVG ANNUAL RENT	8.25% EST. GROSS YIELD
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Off-plan pricing carries a clear premium. Average off-plan price per sq ft is consistently above secondary-market pricing, making the premium measure a central comparison metric.

Luxury communities behave differently from yield leaders. Emirates Hills and Bulgari Resort sit in a high-price/high-rent luxury position, while the highest estimated yields are found in other communities.

High price does not automatically mean high yield. The investment page separates capital-entry price, rental income, rent growth, activity, and estimated yield instead of treating price alone as attractiveness.

Property characteristics show a clear price gradient. Average property price rises strongly with bedroom count, while property type and size help explain segmentation across the market.

Accessibility varies materially by community. Metro travel time and Burj Khalifa distance provide a location lens; The Valley stands out as a high-travel-time observation rather than being removed as an error.

2026 requires partial-year interpretation. Listing and YoY comparisons explicitly account for data availability through April 2026 rather than comparing a partial year with a full prior year.

Community Yield Interpretation

Community yield rankings are aggregate screening indicators based on rental and secondary-market averages under the active filter context, not realized matched-property returns. After Dim Community was expanded to all 84 communities, the Investment & Yield page evaluates the complete community set without grouping unmatched records under a blank member; rankings should therefore be read from the current report and slicer context.

5. Limitations & Professional Considerations

- Listing-level records are synthetic/modelled and calibrated to real market/geographic anchors; results are analytical demonstrations rather than official transaction statistics.
- Estimated gross rental yield combines rental and secondary-market averages under the same filter context; properties are not individually matched.
- 2026 coverage is partial through April, so annual comparisons must be interpreted as year-to-date where relevant.
- Financing-rate analysis uses a shorter common window (Apr 2020-Mar 2024) than the full market dataset.
- Some source fields contain meaningful missingness, notably floor and total-floor information in Secondary Sales (~36% empty).

Data Source & Attribution

Source dataset: *Dubai Real Estate: Sales, Off-Plan & Rentals (2020–2026)* by Kaggle user [sergionefedov](#), licensed under Apache License 2.0. The Power Query transformations, dimensional model, DAX measures, dashboard design, analysis, and portfolio documentation were developed independently for this project.

Source link: <https://www.kaggle.com/datasets/sergionefedov/dubai-real-estate-sales-and-rentals-20202026>

Skills Demonstrated

Capability	Evidence in Project
Power Query	profiling, cleaning, transformation, standardization, dependency-aware model preparation
Data Modeling	fact/dimension design, conformed dimensions, relationship cardinality/filter direction, shared bedroom dimension, referential-integrity QA
DAX	business measures, context-aware comparisons, time intelligence, same-period YoY logic
Visualization	KPI design, trend analysis, scatter analysis, conditional formatting, decomposition tree
Model Optimization	semantic-layer cleanup, hiding redundant technical fields, removing duplicate/debug measures
Business Analysis	translating real-estate questions into interpretable metrics and investment-oriented insights

Outcome

The final solution is a concise, analysis-ready Power BI portfolio project with a clean semantic model, reusable DAX layer, five business-focused report pages, and documented assumptions. It demonstrates the complete workflow from raw-data preparation through dimensional modeling, analytical calculation, visualization, and business interpretation.